# Extremely projective normed spaces

## A characterization in all dimensions

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# Plan

1. Lifting problems
2. Basic theory
3. Dimension restrictions
4. Countable dimension
5. Extremely projective spaces which are not free
6. Uncountable dimension
7. Dimension $\aleph_1$
8. Summary

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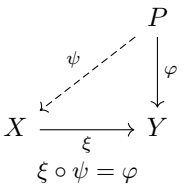
8. Summary

# Lifting problems

Given a "surjection" $\xi\colon X \to Y$ and a map $\varphi\colon P \to Y$: does $\varphi$ lift to $X$?

$P$ is projective if every such problem has a solution $\psi$ with $\|\psi\| \le cC\|\varphi\|$.

Which surjections? In $\mathbf{Nor}$, the category of *normed* spaces and bounded operators, the right notion is quantitative.




## Definition

$\xi\colon X \to Y$ is $c$-topologically surjective ($c \ge 1$) if $c\,\xi(B_X^\circ) \supset B_Y^\circ$, i.e.

$$\forall y \in Y \quad \forall \delta > 0 \quad \exists x \in X : \quad \xi(x) = y, \qquad \|x\| < c\|y\| + \delta.$$

Coisometric: contractive and 1-topologically surjective, $\xi(B_X^\circ) = B_Y^\circ$.

# Topological and extreme projectivity

## Definition

A normed space $P$ is $C$-topologically projective ($C \geq 1$) if for every $c \geq 1$, every $c$-topologically surjective $\xi\colon X \to Y$ between normed spaces and every bounded $\varphi\colon P \to Y$ there is a bounded $\psi\colon P \to X$ with

$$\xi\psi = \varphi, \qquad \|\psi\| \leq cC\|\varphi\|.$$

$P$ is topologically projective if this holds for some $C$, and extremely projective if it holds for *every* $C > 1$.

- $X$ and $Y$ are arbitrary *normed* spaces. Restricting to Banach spaces gives the (different!) theory in $\mathbf{Ban}$.
- Normed spaces = normed modules over the zero algebra. This is the simplest case of Helemskii's *extreme projectivity* for normed modules; his definition looks different but agrees with this one (later).
- Extreme projectivity sits between the metric theory ($C = 1$) and the topological theory (some $C$).

# Banach spaces: the classical answers

**Theorem (Köthe 1966)**

*A Banach space is topologically projective in* $\mathbf{Ban}$ $\iff$ *it is topologically isomorphic to* $\ell_1(\Lambda)$ *for some set* $\Lambda$.

**Theorem (Grothendieck 1955)**

*A Banach space has the metric lifting property* $\iff$ *it is isometrically isomorphic to some* $\ell_1(\Lambda)$.

Both proofs use completeness: open mapping, Pełczyński's decomposition method.

**In $\mathbf{Nor}$ the free objects are different**

$$\ell_1^0(\Lambda) = \{\text{finitely supported families on } \Lambda\}, \qquad \|x\| = \sum_\lambda |x_\lambda|.$$

Bounded families $(x_\lambda)_{\lambda\in\Lambda}$ in a *normed* space $E \leftrightarrow$ bounded operators $T\colon \ell_1^0(\Lambda) \to E$, $T(e_\lambda) = x_\lambda$, $\|T\| = \sup_\lambda \|x_\lambda\|$. No completeness of $E$ is needed, in contrast with $\ell_1(\Lambda)$.

# Normed spaces: what was known

**Theorem (Grønbæk 2016)**

*$E$ is topologically projective in* $\mathbf{Nor}$ $\iff$ *$E$ is topologically isomorphic to* $\ell_1^0(\Lambda)$, $|\Lambda| = \operatorname{dens}(E)$.

**Theorem (Helemskii 2013)**

*$E$ is metrically projective in* $\mathbf{Nor}$ $\iff$ *$E$ is isometrically isomorphic to some* $\ell_1^0(\Lambda)$.

**Extreme projectivity** (Helemskii 2013): $\ell_1^0(\Lambda)$ is extremely projective; $P$ extremely projective $\iff$ $P$ is a near-retract of some $\ell_1^0(\Lambda)$; $P$ extremely projective $\Rightarrow \overline{P} \cong \ell_1(\Lambda)$ isometrically, so $P$ is a dense subspace of some $\ell_1(\Lambda)$; $P$ separable and topologically projective $\Rightarrow \dim P \leq \aleph_0$.

**Question (Helemskii 2013; Grønbæk 2016, Conjecture 3.1)**

Is every extremely projective normed space isometrically isomorphic to some $\ell_1^0(\Lambda)$? More precisely: *which dense subspaces of $\ell_1(\Lambda)$ are extremely projective?*

# The answers, I

## Theorem A (counterexample)

$\ell_1^0(\mathbb{N}) + \mathbb{C}v$, $v = (2^{-n})_n$, is extremely projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$.

## Theorem B (almost isometric characterization)

For a normed space $P$ the following are equivalent:

(i) $P$ is extremely projective;

(ii) $P$ is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some $\Gamma$;

(iv) for every $\varepsilon > 0$ there is $\Lambda$ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

For $\dim P \le \aleph_0$: $P$ extremely projective $\iff$ $\overline{P} \cong \ell_1$ or $\ell_1^n$ isometrically.

So Grønbæk's conjecture fails isometrically and holds almost isometrically.

# The answers, II

## Theorem C (dimension restrictions)

Topologically projective normed spaces contain no infinite dimensional complete subspaces. In particular, no infinite dimensional Banach space is topologically projective in $\mathbf{Nor}$.

## Theorem D (dense subspaces of $\ell_1(\Gamma)$)

A dense $P \subset \ell_1(\Gamma)$ is extremely projective $\iff$ $\Gamma$ has a continuous filtration with countable blocks along which the pieces are dense, form a continuous chain and have countable successive quotients.

## Theorem E (dimension $\aleph_1$)

For $\Gamma = \omega_1$: $\iff$ the pieces $P \cap \ell_1(\delta)$ have countable dimension and the chain jumps only on a non-stationary set. There are $2^{\aleph_1}$ pairwise non-isomorphic dense subspaces of $\ell_1(\omega_1)$ with extremely projective countable pieces, none of them projective.

So for dense subspaces of $\ell_1(\Gamma)$ topological and extreme projectivity coincide, and for uncountable $\Gamma$ this is a genuine restriction beyond Theorem C.

# Why "normed" is the difficulty

Theorem C says: an extremely projective space never contains an infinite dimensional Banach space. So the objects are *incomplete in every direction*, and the usual toolbox is switched off:

- no Baire category, no open mapping, no closed graph;
- no Pełczyński decomposition method ($\ell_1$-sums of the pieces are not inside $P$);
- no Grothendieck-type duality for $P$ itself, only for $\overline{P}$.

**What survives**

The finite dimensional pieces of $P$, the Hamel basis, Hahn–Banach, the Neumann series in the *completion*, and the exact $\ell_1$-norm.

The main tool will be a back-and-forth construction: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ built step by step, which moves $P$ exactly onto $\ell_1^0(\Gamma)$. In the uncountable case the steps are indexed by ordinals and controlled on disjoint blocks of coordinates.

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# Helemskii's definition is the same

Helemskii: $P$ is *extremely projective* if for every *coisometric* $\xi\colon X \to Y$, every $\varphi\colon P \to Y$ and every $\delta > 0$ there is a lifting $\psi$ with $\|\psi\| < \|\varphi\| + \delta$.

## Proposition

*Extremely projective in the sense of the definition above* $\iff$ *extremely projective in the sense of Helemskii.*

## Proof sketch.

($\Rightarrow$) Coisometric operators are 1-topologically surjective; take $C = 1 + \delta/(2\|\varphi\|)$.
($\Leftarrow$) Given a $c$-topologically surjective $\xi$, renorm $X$:

$$\|x\|' = \max\bigl(\|x\|,\ c\|\xi(x)\|\bigr).$$

Then $\xi' = c\xi\colon X' \to Y$ is coisometric: contractive by construction, and $y \in B_Y^\circ$ has a preimage $x \in B_X^\circ$ under $c\xi$ with $\|x\|' = \max(\|x\|, \|y\|) < 1$. A lifting $\psi'$ along $\xi'$ with $\|\psi'\|' < C\|\varphi\|$ gives $\psi = c\psi'$ with $\xi\psi = \varphi$ and $\|\psi\| \le c\|\psi'\|' \le cC\|\varphi\|$. □

# Free spaces are extremely projective

**Proposition**

*$\ell_1^0(\Lambda)$ is $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$.*

**Proof.**

Let $\xi\colon X \to Y$ be $c$-topologically surjective and $\varphi\colon \ell_1^0(\Lambda) \to Y$, $\varphi \neq 0$. For each $\lambda$ choose $x_\lambda \in X$ with

$$\xi(x_\lambda) = \varphi(e_\lambda), \qquad \|x_\lambda\| < c\|\varphi(e_\lambda)\| + c\varepsilon\|\varphi\| \leq c(1+\varepsilon)\|\varphi\|.$$

The family is bounded, so $\psi(e_\lambda) := x_\lambda$ defines a bounded operator with $\|\psi\| = \sup_\lambda \|x_\lambda\| \leq c(1+\varepsilon)\|\varphi\|$ and $\xi\psi = \varphi$ on the Hamel basis. □

**The constant cannot be 1**

The only 1-topologically projective normed space is $\{0\}$: a coisometric functional on $(c_{00}, \|\cdot\|_\infty)$ need not attain its norm (backup slide). So in **Nor** one considers all $C > 1$ at once.

## Retracts and the canonical morphism

$\zeta\colon E \to F$ is a $c$-retraction if $\zeta\eta = 1_F$ for some $\eta$ with $\|\zeta\|\|\eta\| \le c$; then $F$ is a $c$-*retract* of $E$.

**Proposition (permanence)**

*$P'$ $C$-topologically projective, $P$ a $c$-retract of $P' \Rightarrow P$ is $cC$-topologically projective.*
*Hence: a space which is, for every $\varepsilon > 0$, a $(1+\varepsilon)$-retract of an extremely projective space is extremely projective.*

For a normed space $P$ consider the canonical morphism

$$\pi_P\colon \ell_1^0(B_P) \to P, \qquad e_x \mapsto x.$$

Contractive, and $\pi_P(B_{\ell_1^0(B_P)}) = B_P$: strictly coisometric.

**Proposition**

(i) *$P$ $C$-topologically projective $\Rightarrow$ $\pi_P$ has a right inverse $\sigma$ with $\|\sigma\| \le C$ (lift $1_P$ along $\pi_P$);*
(ii) *$\pi_P$ has a right inverse of norm $\le C \Rightarrow P$ is $C(1+\varepsilon)$-topologically projective for all $\varepsilon > 0$.*

One test case suffices: the universal surjection $\pi_P$.

# The criterion

## Theorem

*For a normed space $P$ the following are equivalent:*

(i) *$P$ is extremely projective;*

(ii) *for every $\varepsilon > 0$, $\pi_P$ has a right inverse of norm $\leq 1+\varepsilon$;*

(iii) *for every $\varepsilon > 0$, $P$ is a $(1+\varepsilon)$-retract of some $\ell_1^0(\Lambda)$;*

(iv) *for every $\varepsilon > 0$ there are $\Lambda$, a subspace $Z \subset \ell_1^0(\Lambda)$ and a projection $Q$ of $\ell_1^0(\Lambda)$ onto $Z$ with $\|Q\| \leq 1+\varepsilon$ and $d_{BM}(P, Z) \leq 1+\varepsilon$.*

(i)⇔(iii) is Helemskii's near-retract criterion. (iii)⇒(iv): normalize $\|\zeta\| = 1$, $\|\eta\| \leq 1+\varepsilon$, put $Z = \eta(P)$, $Q = \eta\zeta$. (iv)⇒(i): $\zeta = T^{-1}Q$, $\eta = T$ make $P$ a $(1+\varepsilon)^2$-retract of $\ell_1^0(\Lambda)$; use permanence.

## Takeaway

For every $\varepsilon$, $P$ sits inside a free space $(1+\varepsilon)$-isomorphically and $(1+\varepsilon)$-complemented. We shall construct **isomorphisms onto free spaces** of distortion $< 1+\varepsilon$.

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# Free spaces are small

For infinite $\Lambda$: $\dim \ell_1^0(\Lambda) = |\Lambda| = \operatorname{dens} \ell_1^0(\Lambda)$. An infinite dimensional Banach space has uncountable Hamel dimension (Baire), in fact $\geq \mathfrak{c}$ (Löwig 1934).

## Theorem (hereditary dimension bound)

*Let $P$ be topologically projective and $Z \subset P$ a subspace. Then*

$$\dim Z \leq \max(\operatorname{dens} Z, \aleph_0).$$

*In particular every separable subspace of $P$ has countable dimension.*

## Proof.

Take $\sigma\colon P \to \ell_1^0(B_P)$ with $\pi_P \sigma = 1_P$; $\sigma$ is injective. Let $M \subset Z$ be dense, $|M| = \operatorname{dens} Z$, and $S_0 = \bigcup_{x \in M} \operatorname{supp} \sigma(x)$, so $|S_0| \leq \max(\operatorname{dens} Z, \aleph_0)$. For $z \in Z$, $\sigma(z)$ is a limit of elements supported in $S_0$; coordinate functionals are continuous, so $\sigma(z)$ is supported in $S_0$. Hence $\sigma(Z) \subset \ell_1^0(S_0)$, a space of dimension $|S_0|$. □

Helemskii's argument for separable $P$, in the hereditary form needed later.

# No complete pieces

## Corollary

*Let $P$ be topologically projective, $Z \subset P$, and $\|\cdot\|'$ a norm on $Z$ with $(Z, \|\cdot\|')$ complete and $\|\cdot\| \leq K\|\cdot\|'$. Then $\dim Z < \infty$.*

## Proof.

A separable infinite dimensional $\|\cdot\|'$-closed subspace $Z_0$ would be $\|\cdot\|$-separable with uncountable dimension. □

Every complete subspace of $P$ is finite dimensional. No infinite dimensional Banach space is topologically projective in **Nor**, although $\ell_1(\Lambda)$ is extremely projective in **Ban**.

## Example: completions do not decide

$w_n \geq 1$, $w_n \to \infty$, $P_w = \{x \in \ell_1 : \sum_n w_n|x_n| < \infty\}$ with the $\ell_1$-norm: dense in $\ell_1$, contains $\ell_1^0(\mathbb{N})$, so $\overline{P_w} = \ell_1$; every complete subspace is finite dimensional (open mapping in $\ell_1(w)$); not topologically projective, as $P_w$ is separable but $\dim P_w = \dim \ell_1(w) > \aleph_0$.

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## Theorem (Helemskii 2013; Grønbæk 2016)

*$P$ extremely projective* $\Rightarrow \overline{P} \cong \ell_1(\Gamma)$ *isometrically for some set* $\Gamma$.

*Sketch.* By the criterion, $\overline{P}$ is a $(1+\varepsilon)$-retract of the Banach space $\ell_1(B_P)$ for every $\varepsilon$. Then

$$\overline{P}^* \text{ is } (1+\varepsilon)\text{-injective } \forall\varepsilon \;\Rightarrow\; 1\text{-injective} \;\Rightarrow\; \overline{P}^* \cong C(K),\ K \text{ Stonean}$$
$$\Rightarrow\; \overline{P} \cong L_1(\mu) \;\Rightarrow\; \mu \text{ atomic},$$

the last step because $\overline{P}$ has the Schur property as a complemented subspace of $\ell_1(B_P)$ (Grothendieck; Lacey, Thm. 3.11.6 and §17).

## Two sides of the picture

An extremely projective normed space is, up to isometry,

a **dense subspace of** $\ell_1(\Gamma)$ whose **separable subspaces have countable dimension**.

**Question.** Is every such subspace extremely projective?
Countable $\Gamma$: ✓, and more. Uncountable $\Gamma$: ×, but a correction works.

**Theorem (back-and-forth)**

*Let $D \subset \ell_1$ be a dense subspace of countable dimension and $\varepsilon > 0$. There is a linear bijection $T$ of $\ell_1$ with*

$$\|T\|\|T^{-1}\| \le 1+\varepsilon, \qquad T(D) = \ell_1^0(\mathbb{N}).$$

*In particular $d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.*

**Plan.** $(d_j)$ a Hamel basis of $D$; $\theta_m \in (0, 1/2]$ with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Build automorphisms $F_1, F_2, \dots$ of $\ell_1$ with $\|F_m - 1\| \le \theta_m$ and put $A_j = F_{2j} F_{2j-1} A_{j-1}$, $T = \lim A_j$.

forth $(F_{2j-1})$: make $A_{j-1}(d_j)$ finitely supported;

back $(F_{2j})$: make $e_j$ an element of $A_j(D)$;

while **fixing** a growing finite dimensional $Y_{j-1} \subset \ell_1^0(\mathbb{N})$ that records what was already achieved.

Three lemmas: a *one-vector* lemma, a *hitting* lemma, and a *product* lemma.

# The one-vector lemma

Notation: $x|_N = x$ truncated after coordinate $N$.

## Lemma

*Let $v \in \ell_1$, $N \geq 0$, $0 < \delta < 1$. There is $R\colon \ell_1 \to \ell_1$, $\|R\| \leq \delta$, with*

(i) $R(x) = 0$ *whenever* $x = x|_N$;

(ii) $R(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$;

(iii) $(1 - R)(v) \in \ell_1^0(\mathbb{N})$;

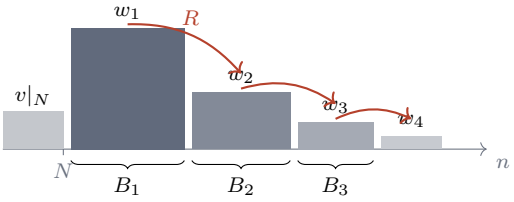
(iv) $(1 - R)^{-1}(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

*Hence $T = 1 - R$ is an automorphism of $\ell_1$, $\|T\| \leq 1 + \delta$, $\|T^{-1}\| \leq (1 - \delta)^{-1}$, $T$ fixes everything supported in $[1, N]$, $T(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$ and*

$$T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) = \ell_1^0(\mathbb{N}).$$

One small perturbation of the identity swallows one vector into $\ell_1^0(\mathbb{N})$, without disturbing a prescribed finite block of coordinates and without letting anything else out of $\ell_1^0(\mathbb{N})$ except multiples of $v$.

# Proof: shift along blocks




Cut the tail $v - v|_N$ into finite blocks $w_k = v|_{B_k}$ with

$$\|w_{k+1}\| \le \delta \|w_k\|.$$

Possible since $t_M = \sum_{n>M} |v_n| \downarrow 0$.

$g_k(x) = \sum_{n \in B_k} \overline{\operatorname{sgn} v_n}\, x_n$: then
$g_k(w_j) = \delta_{jk}\|w_k\|$, $|g_k(x)| \le \sum_{B_k} |x_n|$.

$$R(x) = \sum_{k \ge 1} \frac{g_k(x)}{\|w_k\|}\, w_{k+1}, \qquad \|R\| \le \delta.$$

- $R(w_k) = w_{k+1}$: $R$ shifts the blocks one step to the right.
- $R(v) = \sum_k w_{k+1} = v - v|_N - w_1$, so $(1-R)(v) = v|_N + w_1 \in \ell_1^0(\mathbb{N})$.
- For $x \in \ell_1^0(\mathbb{N})$: $R^m(x) = \sum_{k \in K} c_k w_{k+m}$ (finite $K$), so $(1-R)^{-1}x = \sum_m R^m x = x + \sum_{k \in K} c_k (v - v|_{N_k}) \in \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

## Lemma (hitting)

*Let $Y \subset \ell_1^0(\mathbb{N})$ be finite dimensional and $e_n \notin Y$. There is $C \geq 1$ such that for every $z \in \ell_1^0(\mathbb{N})$ with $\|z - e_n\| \leq (4C)^{-1}$ there is an automorphism $S$ of $\ell_1$ with*

$$\|S - 1\| \leq 2C\|z - e_n\| \leq \tfrac{1}{2}, \qquad S|_Y = 1_Y, \qquad S(z) = e_n.$$

*Proof.* Hahn–Banach: $f \in \ell_1^*$, $f|_Y = 0$, $f(e_n) = 1$, $\|f\| = d^{-1}$, $d = \operatorname{dist}(e_n, Y)$; $C = \max(1, d^{-1})$. Then $|f(z)| \geq 3/4$ and

$$S(x) = x + \frac{f(x)}{f(z)}\,(e_n - z)$$

does the job; it is invertible by the Neumann series. □

A finite dimensional obstacle $Y$ is avoided by a rank-one correction; the closer $z$ is to $e_n$, the smaller the correction. Density of $D$ will supply $z$.

# The product lemma

## Lemma (product)

*$X$ Banach, $\theta_m = \|F_m - 1_X\| < 1$, $\sum_m \theta_m < \infty$, $A_j = F_j \cdots F_1$. Then $A_j \to T$ in norm, $T$ is an automorphism, and*

$$\|T\| \le \prod_m (1+\theta_m), \qquad \|T^{-1}\| \le \prod_m (1-\theta_m)^{-1}.$$

Completeness is used here, in $\ell_1 = \overline{D}$, never in $D$.

Choose the budgets $\theta_m \in (0, 1/2]$ in advance with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Each step may spend at most $\theta_m$ on its perturbation; the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S-1\| \le 2C\|z - e_n\|$) respect any budget.

# The back-and-forth step

Invariants for $A_j$ and the finite dimensional $Y_j \subset \ell_1^0(\mathbb{N})$:

(a) $A_j(d_i) \in Y_j$ for $i \le j$; (forth: basis vectors are caught)

(b) $e_i \in Y_j$ for $i \le j$; (back: unit vectors are caught)

(c) $Y_j \subset A_j(D)$; (everything caught comes from $D$)

(d) $F_{2j-1}$, $F_{2j}$ fix $Y_{j-1}$. (nothing caught is lost)

**Forth.** $N \ge$ all supports of $Y_{j-1}$. One-vector lemma for $v = A_{j-1}(d_j)$, $N$, $\delta = \theta_{2j-1}$: $F_{2j-1} = 1 - R$ fixes $Y_{j-1}$, and $y_j = F_{2j-1}A_{j-1}(d_j) \in \ell_1^0(\mathbb{N})$. Put $A' = F_{2j-1}A_{j-1}$, $Y' = Y_{j-1} + \mathbb{C}y_j \subset A'(D)$.

**Back.** If $e_j \in Y'$, done. Otherwise $A'(D)$ is dense, so pick $d \in D$ with $\|A'(d) - e_j\| \le \theta_{2j}/(16C)$.

- one-vector lemma for $A'(d)$ (fixing $Y'$ and $e_j$), $\delta = \theta_{2j}/2$: $G = 1 - R'$ with $z = GA'(d) \in \ell_1^0(\mathbb{N})$, $\|z - e_j\| \le (4C)^{-1}$;
- hitting lemma: $S$ fixes $Y'$, $S(z) = e_j$, $\|S - 1\| \le 5\theta_{2j}/32$.

$F_{2j} = SG$, $\|F_{2j} - 1\| \le \theta_{2j}$, $Y_j = Y' + \mathbb{C}e_j$, and $e_j = A_j(d) \in A_j(D)$.

## Passing to the limit

By the product lemma $A_j \to T$, an automorphism of $\ell_1$ with

$$\|T\|\|T^{-1}\| \le \prod_m \frac{1+\theta_m}{1-\theta_m} \le 1+\varepsilon.$$

- $T(D) \subset \ell_1^0(\mathbb{N})$: $A_i(d_i) \in Y_i$ by (a), and all later $F_m$ fix $Y_i$ by (d), so $T(d_i) = A_i(d_i) \in \ell_1^0(\mathbb{N})$.
- $\ell_1^0(\mathbb{N}) \subset T(D)$: by (b), (c) there is $d \in D$ with $A_i(d) = e_i \in Y_i$, and again $T(d) = A_i(d) = e_i$.

So $T(D) = \ell_1^0(\mathbb{N})$. □

### What made it work

- each step is a perturbation of the identity of size $\theta_m$, and $\sum \theta_m$ is as small as we like: the constants are *kept*;
- each step is *local*: it fixes a finite block of coordinates and acts on the tail; the one-vector lemma produces nothing outside $\ell_1^0(\mathbb{N}) + \mathbb{C}v$;
- the Hamel basis is countable, so a single $\omega$-sequence of steps exhausts $D$.

# Countable dimension: the characterization

## Theorem

*Let* $\dim P \leq \aleph_0$. *The following are equivalent:*

(i) $P$ *is extremely projective;*

(ii) $\overline{P} \cong \ell_1(\Gamma)$ *isometrically,* $\Gamma$ *countable;*

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ *for some countable* $\Gamma$*;*

(iv) *for every* $\varepsilon > 0$ *there is* $\Lambda$ *with* $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

*Proof.* (i)⇒(ii): completion theorem; $\overline{P}$ is separable, so $\Gamma$ is countable. (ii)⇒(iii): $J(P) \subset \ell_1$ is dense and of countable dimension; back-and-forth. (iv)⇒(i): a $(1+\varepsilon)$-isomorphism onto $\ell_1^0(\Lambda)$ is a $(1+\varepsilon)$-retraction; criterion. □

## Corollary

*Every dense subspace of* $\ell_1$ *of countable dimension is extremely projective and has Banach–Mazur distance* $1$ *to* $\ell_1^0(\mathbb{N})$.

No projectivity assumption in (ii): extreme projectivity *is* a property of the completion.

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# Extreme points

By the corollary, $\ell_1^0(\mathbb{N}) + V$ is extremely projective for every $V \subset \ell_1$ of countable dimension. Is it ever free?

## Lemma

*Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the $\ell_1$-norm. Then*

$$\operatorname{ext}(B_E) = \{\lambda e_s : \ s \in S,\ |\lambda| = 1\}, \qquad \operatorname{conv}(\operatorname{ext} B_E) = B_{\ell_1^0(S)}.$$

## Proof idea.

The $\lambda e_s$ are extreme in $B_{\ell_1(S)}$, hence in $B_E = B_{\ell_1(S)} \cap E$. If $x \in E$, $\|x\| = 1$, has two non-zero coordinates $x_n, x_m$, put $t = \min(|x_n|, |x_m|)$ and

$$u = t\big(\operatorname{sgn}(x_n)e_n - \operatorname{sgn}(x_m)e_m\big) \in \ell_1^0(S) \subset E.$$

Then $\|x \pm u\| = (|x_n| \pm t) + (|x_m| \mp t) + \cdots = 1$, so $x$ is not extreme. Finite convex combinations of the $\lambda e_s$ are exactly the elements of $B_{\ell_1^0(S)}$. □

The lemma only sees finitely supported vectors: $E$ may be any space between $\ell_1^0(S)$ and $\ell_1(S)$.

# The counterexample

**Theorem**

*Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the $\ell_1$-norm. If $E \cong \ell_1^0(\Lambda)$ isometrically, then $E = \ell_1^0(S)$.*

**Proof.**

An isometry $T\colon \ell_1^0(\Lambda) \to E$ maps extreme points onto extreme points, so

$$B_E = T\big(\operatorname{conv}\operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\big(T \operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\operatorname{ext} B_E = B_{\ell_1^0(S)}.$$

□

**Corollary (answer to Helemskii and Grønbæk)**

For $V \subset \ell_1$ of countable dimension with $V \not\subset \ell_1^0(\mathbb{N})$, e.g. $V = \mathbb{C}v$, $v = (2^{-n})$, the space $\ell_1^0(\mathbb{N}) + V$ is extremely projective and **not** isometrically isomorphic to any free normed space.

# Between $\ell_1^0(\mathbb{N})$ and $\ell_1$

| space | extremely projective | isometric to a free space |
|---|---|---|
| $\ell_1^0(\mathbb{N})$ | ✓ | ✓ |
| $\ell_1^0(\mathbb{N}) + V$, $\dim V \leq \aleph_0$, $V \not\subset \ell_1^0(\mathbb{N})$ | ✓ | × |
| $P_w$ (weighted) | × | × |

All three have completion $\ell_1$: the difference is invisible on completions. By the theorem, *no* space strictly between $\ell_1^0(\mathbb{N})$ and $\ell_1$ is free, yet by the corollary all those of countable dimension are extremely projective.

The next section gives a criterion which distinguishes the projective ones among all dense subspaces, in any dimension.

# Plan

1. Lifting problems

2. Basic theory

3. Dimension restrictions

4. Countable dimension

5. Extremely projective spaces which are not free

6. Uncountable dimension

7. Dimension $\aleph_1$

8. Summary

## What changes

Countable dimension: dense in $\ell_1$ + countable dimension $\Rightarrow$ extremely projective.

**Hope.** Dense in $\ell_1(\Gamma)$ + all separable subspaces of countable dimension $\Rightarrow$ extremely projective?

**No.** Already for $\Gamma = \omega_1$ (ladder spaces, Section 7).

### What is true (Theorem 6.7)

A dense $P \subset \ell_1(\Gamma)$ is extremely projective $\iff$ $\Gamma$ has a continuous filtration with **countable blocks** along which the pieces $P \cap \ell_1(\Gamma_\delta)$ are dense, form a **continuous chain**, and have **countable dimensional successive quotients**.

Two tools:

- a transfinite back-and-forth along such a filtration, each countable block solved by the countable theorem, perturbations confined to disjoint blocks;
- closed pairs, an elementary-submodel argument producing such a filtration for every projective $P$.

## Filtrations and blocks

A filtration of $\Gamma$: an increasing family $(\Gamma_\delta)_{\delta\le\lambda}$, $\Gamma_0=\varnothing$, $\Gamma_\lambda=\Gamma$, continuous at limits. Blocks $N_\delta=\Gamma_{\delta+1}\setminus\Gamma_\delta$, so $\Gamma_\delta=\bigsqcup_{\delta'<\delta}N_{\delta'}$ and $\ell_1(\Gamma_\delta)=\big(\bigoplus_{\delta'<\delta}\ell_1(N_{\delta'})\big)_{\ell_1}$.

For $P\subset\ell_1(\Gamma)$: $P_\delta=P\cap\ell_1(\Gamma_\delta)$ and

$$D_\delta=\{x|_{N_\delta}:x\in P_{\delta+1}\}\subset\ell_1(N_\delta),\qquad D_\delta\cong P_{\delta+1}/P_\delta.$$

**Theorem (transfinite back-and-forth)**

*Let $P$ be dense in $\ell_1(\Gamma)$ and $(\Gamma_\delta)$ a filtration with countable blocks such that*

(i) *$P_\delta$ is dense in $\ell_1(\Gamma_\delta)$ for all $\delta\le\lambda$;*

(ii) *$P_\delta=\bigcup_{\delta'<\delta}P_{\delta'}$ for limit $\delta\le\lambda$* *(continuity);*

(iii) *$\dim P_{\delta+1}/P_\delta\le\aleph_0$ for all $\delta<\lambda$* *(small blocks).*

*Then for every $\varepsilon>0$ there is an automorphism $T$ of $\ell_1(\Gamma)$ with $T(P)=\ell_1^0(\Gamma)$, $\|T\|\le1+\varepsilon$, $\|T^{-1}\|\le1$.*

Each block is solved first: $D_\delta$ is dense in $\ell_1(N_\delta)$ and of countable dimension, so the countable theorem gives $U_\delta$ with $U_\delta(D_\delta)=\ell_1^0(N_\delta)$, $\|U_\delta\|\le1+\theta$, $\|U_\delta^{-1}\|\le1$.

# The shape of $T$

Recursively choose $V_\delta\colon \ell_1(N_\delta) \to \ell_1(\Gamma_\delta)$ with $\|V_\delta\| \le \theta$, and put for $x \in \ell_1(\Gamma_\delta)$

$$W_\delta(x) = x + \sum_{\delta'<\delta} V_{\delta'}(x|_{N_{\delta'}}), \qquad U_{<\delta} = \bigoplus_{\delta'<\delta} U_{\delta'}, \qquad T_\delta = U_{<\delta} W_\delta.$$

- The $N_{\delta'}$ are disjoint and $\|x\| = \sum_{\delta'} \|x|_{N_{\delta'}}\|$, so $\|W_\delta - 1\| \le \theta$ regardless of $\delta$: no product of $\lambda$ factors ever appears.
- $\|U_{<\delta}\| \le 1+\theta$, $\|U_{<\delta}^{-1}\| \le 1$ ($\ell_1$-sum).
- $T_\delta$ extends $T_{\delta'}$ for $\delta' < \delta$.

## Inductive claim

$$T_\delta(P_\delta) = \ell_1^0(\Gamma_\delta) \qquad \text{for all } \delta \le \lambda.$$

Limit stages are free: by (ii) and compatibility, $T_\mu(P_\mu) = \bigcup_{\delta<\mu} T_\delta(P_\delta) = \bigcup_{\delta<\mu} \ell_1^0(\Gamma_\delta) = \ell_1^0(\Gamma_\mu)$.

Compare with the countable proof: there $T$ was an infinite product of perturbations and the limit stage needed the product lemma. Here the perturbations live on disjoint blocks and *add* instead of multiplying.

# The successor step

Let $U = U_\delta$, $y_n = U^{-1}(e_n)$ for $n \in N_\delta$: a Hamel basis of $D_\delta$ with $\|\sum c_n y_n\| \geq (1+\theta)^{-1} \sum |c_n|$.

- Lift: $x_n \in P_{\delta+1}$ with $x_n|_{N_\delta} = y_n$.
- Correct: $P_\delta$ is dense in $\ell_1(\Gamma_\delta)$ by (i), so there is $z_n \in \ell_1(\Gamma_\delta)$ with

$$x_n|_{\Gamma_\delta} + z_n \in P_\delta, \qquad \|z_n\| \leq \theta(1+\theta)^{-2}.$$

- Define $V_\delta(\sum c_n y_n) = \sum c_n W_\delta(z_n)$ on $D_\delta$; then $\|V_\delta\| \leq \theta$, extend to $\ell_1(N_\delta)$.

## Key inclusion

$$x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta}) \in P_\delta \qquad \text{for every } x \in P_{\delta+1}.$$

($x|_{N_\delta} = \sum c_n y_n$; $x - \sum c_n x_n$ vanishes on $N_\delta$, hence lies in $P_\delta$.)

Then for $x \in P_{\delta+1}$:

$$T_{\delta+1}(x) = T_\delta\big(x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta})\big) + U(x|_{N_\delta}) \in \ell_1^0(\Gamma_\delta) + \ell_1^0(N_\delta),$$

and $T_{\delta+1}(x_n) \in \ell_1^0(\Gamma_\delta) + e_n$ gives the reverse inclusion. □

# Where do good filtrations come from?

Fix a dense topologically projective $P \subset \ell_1(\Gamma)$, a retraction $\sigma\colon P \to \ell_1^0(S)$, $\pi\colon \ell_1^0(S) \to P$, $\pi\sigma = 1_P$, $\|\pi\| \le 1$, and sequences $p_n^\gamma \in P$ with $p_n^\gamma \to e_\gamma$. $P_A = P \cap \ell_1(A)$.

## Definition

$(A, B)$, $A \subset \Gamma$, $B \subset S$, is closed if

(a) $\operatorname{supp} \pi(e_s) \subset A$ for $s \in B$;

(b) $\operatorname{supp} \sigma(x) \subset B$ for $x \in P_A$;

(c) $\operatorname{supp} p_n^\gamma \subset A$ for $\gamma \in A$, $n \in \mathbb{N}$.

## Lemma

*For a closed pair: $P_A = \pi(\ell_1^0(B))$, $P_A$ is dense in $\ell_1(A)$, and $\sigma|_{P_A}$, $\pi|_{\ell_1^0(B)}$ exhibit $P_A$ as a retract of $\ell_1^0(B)$. So $P_A$ is topologically projective.*

*Proof.* (a): $\pi(\ell_1^0(B)) \subset P_A$. (b): $x = \pi\sigma(x) \in \pi(\ell_1^0(B))$. (c): $\overline{P_A} \ni e_\gamma$ for $\gamma \in A$. □

Closed pairs are the "elementary submodels" of the situation.

# Countable quotients and countable closing off

## Lemma (countable quotient)

*$(A, B)$ closed, $A' \supset A$ with $A' \setminus A$ countable. Then* $\dim P_{A'}/P_A \leq \aleph_0$.

*Proof.* $D = \{x|_{A'\setminus A} : x \in P_{A'}\} \cong P_{A'}/P_A$ is separable. The map $\rho(y) = \sigma(x)|_{S\setminus B}$ ($x \in P_{A'}$, $x|_{A'\setminus A} = y$) is well defined, bounded and **injective**: $\rho(y) = 0$ forces $\sigma(x) \in \ell_1^0(B)$, so $x = \pi\sigma(x) \in P_A$ and $y = 0$. A bounded injection into a free space bounds the dimension by the density (the argument of the dimension bound). □

## Lemma (countable closing off)

*$(A, B)$ closed, $N \subset \Gamma$ countable. There is a closed pair $(A', B') \supset (A \cup N, B)$ with $A' \setminus A$, $B' \setminus B$ countable.*

*Proof.* Alternate $\omega$ times: add the supports of $\sigma(x)$ for $x$ in a basis of $P_{A_k}$ *modulo* $P_A$ (countable by the first lemma), then supports of $\pi(e_s)$ and of the $p_n^\gamma$ for the new points. Unions. (a), (c) are clear; (b) by density: $\bigcup_k P_{A_k}$ is dense in $\ell_1(A')$, and $\sigma$ is continuous. □

Subtlety: $\operatorname{supp} x$ need not sit in one $A_k$, so density, not a support argument, proves (b).

## Lemma (unions)

*The union of an increasing chain $((A_\delta, B_\delta))_{\delta<\lambda}$ of closed pairs is a closed pair, and*

$$P_{A_\lambda} = \pi(\ell_1^0(B_\lambda)) = \bigcup_{\delta<\lambda} \pi(\ell_1^0(B_\delta)) = \bigcup_{\delta<\lambda} P_{A_\delta}.$$

*Proof.* (a), (c) are inherited. (b): $\bigcup_\delta P_{A_\delta}$ is dense in $\ell_1(A_\lambda)$ (each $P_{A_\delta}$ is dense in $\ell_1(A_\delta)$), and the same continuity argument gives $\sigma(x) \in \ell_1^0(B_\lambda)$ for $x \in P_{A_\lambda}$. The formula for $P_{A_\lambda}$ follows from $P_A = \pi(\ell_1^0(B))$ for closed pairs. □

Continuity (ii) is **automatic** along chains of closed pairs, for any cofinality. This is exactly what the ladder spaces of the next section lack: there is no retraction $\sigma$, and the chain jumps on a stationary set.

# The criterion

**Theorem**

*Let $P$ be a dense subspace of $\ell_1(\Gamma)$. The following are equivalent:*

(i) *$P$ is extremely projective;*

(ii) *$P$ is topologically projective;*

(iii) *$\Gamma$ has a filtration with countable blocks satisfying (a) dense pieces, (b) continuity, (c) countable successive quotients.*

*If so, for every $\varepsilon > 0$ there is an automorphism $T$ of $\ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$, $\|T\| \leq 1 + \varepsilon$, $\|T^{-1}\| \leq 1$; in particular $d_{BM}(P, \ell_1^0(\Gamma)) = 1$.*

*Proof.* (ii)⇒(iii): enumerate $\Gamma = \{\gamma_\alpha\}_{\alpha<\kappa}$ and build closed pairs: $A_0 = B_0 = \varnothing$; close off $A_\alpha \cup \{\gamma_\alpha\}$ at successors (countable increment); unions at limits. (a) closed pairs are dense; (b) unions lemma; (c) countable quotient lemma. (iii)⇒(i): the transfinite theorem gives $T$; then $T|_P$ and $T^{-1}|_{\ell_1^0(\Gamma)}$ make $P$ a $(1+\varepsilon)$-retract of $\ell_1^0(\Gamma)$; the criterion of Section 2. □

One transfinite construction with countable steps; no induction on cardinals.

# The characterization in all dimensions

## Theorem

*For a normed space $P$ the following are equivalent:*

(i) *$P$ is extremely projective;*

(ii) *$P$ is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;*

(iii) *$d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some set $\Gamma$;*

(iv) *for every $\varepsilon > 0$ there is $\Lambda$ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.*

*In (iii), $|\Gamma| = \dim P$ for infinite dimensional $P$, and the distance is realized by bijections $T\colon \overline{P} \to \ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\|$ arbitrarily close to $1$.*

*Proof.* (i)⇒(ii): completion theorem. (ii)⇒(iii): $J(P)$ is a dense topologically projective subspace of $\ell_1(\Gamma)$; general theorem. (iv)⇒(i): criterion. □

## Grønbæk's conjecture

Every extremely projective normed space is **almost isometric** to $\ell_1^0(\Gamma)$, $|\Gamma| = \dim P$. The conjecture holds in the almost isometric form and fails in the isometric form.

# Helemskii's question on dense subspaces

**Which dense subspaces $P \subset \ell_1(\Lambda)$ are extremely projective?**

## Answer

$P$ extremely projective $\iff$ $P$ topologically projective $\iff$ $P$ admits a filtration with countable blocks, dense pieces, continuity and countable quotients.

- $\Lambda$ countable: $\iff \dim P \leq \aleph_0$.
- $\Lambda$ uncountable: a genuine restriction beyond dimension; the continuity condition fails for the ladder spaces of the next section.
- The answer refers only to the subspaces $P \cap \ell_1(A)$, and the coordinates of $\ell_1(\Lambda)$ are intrinsic: the extreme points of the unit ball are the $\lambda e_\gamma$ (Section 5). No retraction onto a free space is mentioned.
- Topological and extreme projectivity coincide for dense subspaces of $\ell_1(\Gamma)$. This is a feature of the *exact* norm: if $E$ is $\ell_1(\Gamma)$ with an equivalent, not almost isometric, norm, then $J(\ell_1^0(\Gamma)) \subset E$ is topologically but not extremely projective.

# Plan

1. Lifting problems

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8. Summary

# Set theory in one slide

- $\omega_1$ = the first uncountable ordinal; its elements are the countable ordinals. Every $x \in \ell_1(\omega_1)$ has countable support, hence lives in some $\ell_1(\delta)$, $\delta < \omega_1$.
- $C \subset \omega_1$ is closed unbounded (club) if unbounded and closed under countable suprema. The limit ordinals form a club; two clubs meet in a club; for $h\colon \omega_1 \to \omega_1$ the ordinals closed under $h$ form a club.
- $E \subset \omega_1$ is stationary if it meets every club; *non-stationary* = disjoint from some club = "small".
- **Solovay.** $\omega_1$ is the union of $\aleph_1$ pairwise disjoint stationary sets.

## The objects

$P$ dense in $\ell_1(\omega_1)$, $P_\delta = P \cap \ell_1(\delta)$ for $\delta < \omega_1$: an increasing $\omega_1$-chain of subspaces with union $P$.

All that matters about an $\omega_1$-chain is what happens on a club. The enumeration of the index set changes the chain only off a club.

# The invariant and the criterion

## Definition

$\Delta(P) = \{\delta < \omega_1 \text{ limit} : \ P_\delta \neq \bigcup_{\delta' < \delta} P_{\delta'}\}$, the set of jumps of the chain.

## Theorem

*A dense $P \subset \ell_1(\omega_1)$ is extremely projective* $\iff$

(i) *every $P_\delta$, $\delta < \omega_1$, has countable dimension (equivalently: every separable subspace of $P$ does), and*

(ii) $\Delta(P)$ *is not stationary.*

*Proof.* ($\Rightarrow$) (i) is the dimension bound. For (ii) take the chain of closed pairs $(A_\alpha, B_\alpha)$ from the criterion with $\gamma_\alpha = \alpha$. On a club, $A_\delta = \delta$; there $P_\delta = P_{A_\delta} = \bigcup_{\alpha<\delta} P_{A_\alpha}$ by the unions lemma, and each $A_\alpha$ sits below some $\delta' < \delta$.
($\Leftarrow$) Density of $P_\delta$ holds on a club $C_1$; a club $C_2$ misses $\Delta(P)$. Enumerate $C_1 \cap C_2 \cup \{0\}$ as $(c_\alpha)$ and take $\Gamma_\alpha = c_\alpha$: countable blocks, (a) by $C_1$, (b) by $C_2$ and continuity of the enumeration, (c) by (i). Apply the criterion. □

# Both conditions are needed; the invariant is invariant

- $P_w + \ell_1^0(\omega_1 \setminus \omega)$ ($P_w$ the weighted space on $\omega$): $\Delta(P) = \{\omega\}$, not stationary, but $P_\omega = P_w$ has uncountable dimension. Not projective.
- Ladder spaces on stationary sets (next slide): (i) holds, (ii) fails.

**Proposition**

*If $T$ is a bounded bijection of $\ell_1(\omega_1)$ with $T(P) = P'$, then $\Delta(P) \triangle \Delta(P')$ is not stationary. Hence the class of $\Delta(P)$ modulo non-stationary sets is an isomorphism invariant of the normed space $P$, independent of the enumeration of the index set.*

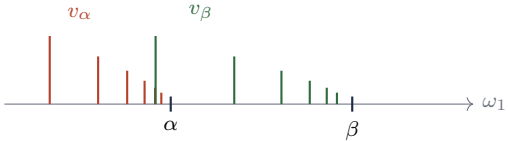
*Proof.* $F(\delta) = \sup_{\beta<\delta}(\sup \operatorname{supp} Te_\beta + 1)$, similarly $G$ for $T^{-1}$. For limit $\delta$ closed under $F$ and $G$ (a club): $T(\ell_1(\delta)) = \ell_1(\delta)$, so $T(P_\delta) = P'_\delta$ and $T\big(\bigcup_{\delta'<\delta} P_{\delta'}\big) = \bigcup_{\delta'<\delta} P'_{\delta'}$; jumps correspond. An isomorphism $P \to P'$ extends to the completions. □

# Ladder spaces

$E$ a set of limit ordinals $< \omega_1$. For $\alpha \in E$ fix a ladder $\alpha_1 < \alpha_2 < \cdots \nearrow \alpha$ and put

$$v_\alpha = \sum_{n \geq 1} 2^{-n} e_{\alpha_n},$$

$$P_E = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha : \alpha \in E\}.$$




Two ladders share only finitely many rungs.

## Theorem

(i) $(P_E)_\delta = \ell_1^0(\delta) + \operatorname{span}\{v_\alpha : \alpha \in E,\ \alpha \leq \delta\}$: *countable dimension, dense, hence* ***extremely projective*** *for every* $\delta < \omega_1$;

(ii) $\Delta(P_E) = E$ ($v_\delta$ *enters at stage* $\delta$ *but is not a finite combination of earlier vectors modulo* $\ell_1^0(\delta)$);

(iii) $P_E$ *is extremely projective* $\iff$ $E$ *is not stationary;*

(iv) $E \triangle E'$ *stationary* $\Rightarrow P_E \ncong P_{E'}$.

# $2^{\aleph_1}$ spaces, and an old theory in new clothes

Solovay: pairwise disjoint stationary $S_\xi \subset \omega_1$, $\xi < \omega_1$. For $X \subset \omega_1$ put $E_X = \bigcup_{\xi \in X} S_\xi$. Then $E_X \triangle E_Y \supset S_\xi$ for $X \neq Y$.

$2^{\aleph_1}$ pairwise non-isomorphic dense subspaces of $\ell_1(\omega_1)$, all with extremely projective countable pieces; only $P_\varnothing = \ell_1^0(\omega_1)$ is extremely projective. All of them contain the same dense subspace $\ell_1^0(\omega_1)$.

| almost free abelian groups | extremely projective normed spaces |
|---|---|
| Pontryagin, Hill: filtrations by countable subgroups with free quotients | Theorem 6.7: filtrations with countable blocks and countable quotients |
| Eklof's invariant $\Gamma(G)$ | $\Delta(P)$ modulo non-stationary sets |
| $\aleph_1$-free non-free groups built from ladder systems | the spaces $P_E$ |

In the group case the picture above $\aleph_1$ depends on set theory; expect the same here.

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1. Lifting problems
2. Basic theory
3. Dimension restrictions
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5. Extremely projective spaces which are not free
6. Uncountable dimension
7. Dimension $\aleph_1$
8. Summary

# Where each ingredient was consumed

| ingredient | used for | where |
|---|---|---|
| universal property of $\ell_1^0(\Lambda)$ | free spaces are extremely projective; canonical morphism $\pi_P$ | §2 |
| retraction $\sigma$ onto a free space | supports are finite: dimension bound, closed pairs, countable quotients | §3, §6 |
| Grothendieck (via duality of $\overline{P}$) | completion is $\ell_1(\Gamma)$ | §4 |
| exact $\ell_1$-norm: disjoint blocks add | $\|R\| \leq \delta$ in the one-vector lemma; $\|W_\delta - 1\| \leq \theta$ transfinitely | §4, §6 |
| Hahn–Banach, Neumann series in $\overline{P}$ | hitting lemma; invertibility | §4 |
| density of $P_\delta$ in $\ell_1(\Gamma_\delta)$ | corrections $z_n$ at successor steps | §6 |
| extreme points of $B_{\ell_1}$ | no space strictly between $\ell_1^0(S)$ and $\ell_1(S)$ is free; coordinates are intrinsic | §5, §7 |
| clubs and stationary sets, Solovay | the $\aleph_1$ criterion; $2^{\aleph_1}$ ladder spaces | §7 |

# Comparison with Grønbæk's proof

Grønbæk: topologically projective $\Rightarrow$ isomorphic to $\ell_1^0(\Lambda)$, in two steps.

1. Transfinite construction: every subspace $Y \subset \ell_1^0(\Lambda)$ contains a complemented copy of $\ell_1^0(\Lambda')$, $|\Lambda'| = \operatorname{dens} Y$.
   Inspection: the copy is $(1+\varepsilon)$-isomorphic and $(1+\varepsilon)$-complemented.
2. Pełczyński's decomposition method.
   Here the constants are lost.

- decomposition method: $Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, then $Y \cong Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, with uncontrolled constants;
- back-and-forth: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ moving $P$ exactly onto $\ell_1^0(\Gamma)$.

The price of keeping the constants is the continuity condition (ii), and the closed pairs show that it is always available for projective spaces.

# Open problems

- Dimension $\geq \aleph_2$. Can the filtration condition be replaced by a condition on the pieces $P \cap \ell_1(A)$ for $A$ in a club of subsets of $\Gamma$? Is there a dense subspace of $\ell_1(\omega_2)$ which is not projective although every piece of size $\aleph_1$ is? For groups the answers depend on set theory.
- Dense subspaces of other Banach spaces: for $E \ncong \ell_1(\Gamma)$ almost isometrically there are topologically but not extremely projective dense subspaces. Which $E$ admit extremely projective dense subspaces at all? (Only those with $E \cong \ell_1(\Gamma)$ isometrically.)
- Extreme projectivity over non-zero algebras: Helemskii's original setting of sequence algebras. Is there an analogue of the almost isometric characterization, of the counterexample, of the invariant?

## The real question

In **Nor** the free objects are tiny and the projective objects are almost free. What replaces the Banach space toolbox when every complete subspace is finite dimensional?

# Summary

- **Extremely projective** $= C$-topologically projective for every $C > 1$, in the category of normed spaces.
- **Counterexample**: $\ell_1^0(\mathbb{N}) + \mathbb{C}v$ is extremely projective and not free. The isometric conjecture of Helemskii and Grønbæk fails.
- **Characterization**: extremely projective $\iff$ topologically projective with completion $\ell_1(\Gamma) \iff d_{BM}(P, \ell_1^0(\Gamma)) = 1$. The conjecture holds almost isometrically.
- **Countable dimension**: a property of the completion alone.
- **Uncountable dimension**: filtrations with countable blocks; for $\aleph_1$ a stationary-set invariant and $2^{\aleph_1}$ ladder spaces.
- **Method**: back-and-forth, countable and transfinite, with perturbations on disjoint blocks; closed pairs.

Thank you!

# References


A. Ya. Helemskii, Extreme version of projectivity for normed modules over sequence algebras, Canad. J. Math. 65 (2013), 559–574.

A. Ya. Helemskii, Metric freeness and projectivity for classical and quantum normed modules, Sb. Math. 204 (2013), 1056–1083.

N. Grønbæk, Lifting problems for normed spaces, Ann. Funct. Anal. 7 (2016), 118–126.

G. Köthe, Hebbare lokalkonvexe Räume, Math. Ann. 165 (1966), 181–195.

A. Grothendieck, Une caractérisation vectorielle-métrique des espaces $L_1$, Canad. J. Math. 7 (1955), 552–561.

P. Hájek, T. Russo, On densely isomorphic normed spaces, J. Funct. Anal. 279 (2020), 108667.

J. Ferrer, P. Koszmider, W. Kubiś, Almost disjoint families of countable sets and separable complementation properties, J. Math. Anal. Appl. 401 (2013), 939–949.

P. C. Eklof, A. H. Mekler, Almost free modules. Set-theoretic methods, North-Holland, 2002.

T. Jech, Set theory, the third millennium edition, Springer, 2003.

N. T. Nemesh, Extremely projective normed spaces, preprint.

# Backup: no 1-topologically projective spaces

Let $X = c_{00}$ with the sup-norm, $s = \sum_n 2^{-n}(1-2^{-n})$ and

$$f(x) = s^{-1} \sum_{n \geq 1} 2^{-n}(1-2^{-n})\, x_n, \qquad \|f\| \leq 1.$$

- $f$ is coisometric: for $|z| < 1$ pick $N$ with $|z| < s^{-1} \sum_{n \leq N} 2^{-n}(1-2^{-n})$ and $x = t \operatorname{sgn}(z) \sum_{n \leq N} e_n$ with suitable $t \in [0,1)$; then $f(x) = z$, $\|x\| < 1$.
- $f$ does not attain its norm: $|f(x)| = 1$, $\|x\| \leq 1$ would force $x_n = \operatorname{sgn} f(x)$ for all $n$.

Let $P \neq 0$ be 1-topologically projective, $p \in P$, $\|p\| = 1$, and $g \in P^*$ with $\|g\| = g(p) = 1$ (Hahn–Banach). Lift $g$ along $f$: $\psi\colon P \to X$, $f\psi = g$, $\|\psi\| \leq 1$. Then $x = \psi(p) \in B_X$ and $f(x) = g(p) = 1$. Contradiction. □

The same functional on $\ell_1$, $x \mapsto \sum (1-2^{-n}) x_n$, shows that with the open-ball convention 1-topological projectivity is empty in **Ban** too. With closed balls (strictly coisometric operators) one recovers the metric theory; for extreme projectivity the choice is immaterial.

# Backup: the product lemma

$\theta_m = \|F_m - 1_X\| < 1$, $\sum \theta_m < \infty$, $A_j = F_j \cdots F_1$, $M = \prod(1+\theta_m)$, $M' = \prod(1-\theta_m)^{-1}$.

- $\|A_j\| \le M$ and, by the Neumann series, $\|F_m^{-1}\| \le (1-\theta_m)^{-1}$, so $\|A_j^{-1}\| \le M'$.
- $\|A_j - A_{j-1}\| = \|(F_j - 1)A_{j-1}\| \le \theta_j M$: Cauchy, $A_j \to T$, $\|T\| \le M$.
- $\|A_j^{-1} - A_k^{-1}\| = \|A_j^{-1}(A_k - A_j)A_k^{-1}\| \le {M'}^2 \|A_k - A_j\|$: Cauchy, $A_j^{-1} \to B$, $\|B\| \le M'$.
- $A_j A_j^{-1} = A_j^{-1} A_j = 1$ in the limit: $TB = BT = 1$. □

In the countable back-and-forth the $\theta_m$ are chosen in advance with $\prod(1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$; every step then has a fixed budget $\theta_m$ for its perturbation, and the budget is respected by the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S - 1\| \le 2C\|z - e_n\|$, with $z$ chosen close enough to $e_n$ by density).